

Physiology: Diffusion of Gases Through the Respiratory Membrane

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Objectives

- Describe the principles behind the mechanisms of gas exchange based on the fundamental properties of gases and their behavior in solution. Define and explain the General Gas Law, Boyles Law, Dalton's Law, Henry's Law, and Fick's Law.
- Relate the concept of Lung Diffusion Capacity (DL) to Fick's Law and apply to pathophysiological examples where DL is becoming limited.
- Explain the principles behind gas exchange in the lungs and describe partial pressure changes in PO₂ and PCO₂ along the way.
- Explain the effect of physiologic humidification on tracheal PO₂? Is tracheal PCO₂ affected as well? Why is it or why not?
- Predict the effect of blood shunting through bronchial and coronary venous return on the PO₂ of systemic arterial blood.
- Compare and contrast diffusion-limited with perfusion-limited gas exchange. How do they apply to the respiratory exchange of O₂ and CO₂ under normal physiologic, pathophysiologic and challenge conditions (exercise, high altitude)?

The General Gas Law

$$PV = nRT$$

P = Pressure (mm Hg)

V = Volume (L)

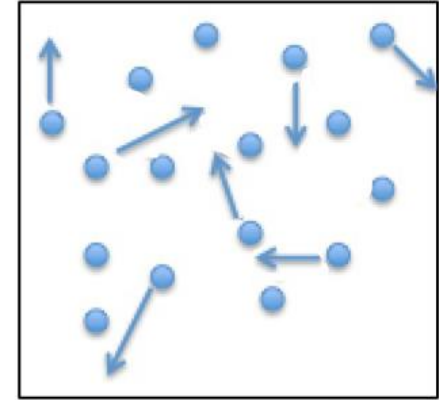
n = Moles (mol)

R = Gas constant ($\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$)

T = Temperature(K)



Propane Gas Tank



Molecules inside the gas tank

We assume:

BTPS: body temperature (37°C , or 310K), ambient pressure, & gas saturated with water vapor

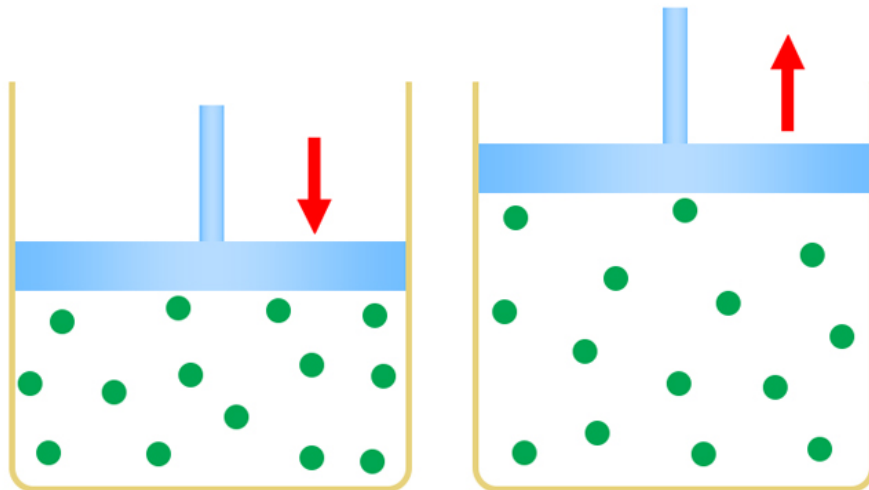
or STPD: standard temperature (0°C , or 273K), standard pressure (760 mmHg), and dry gas

Boyle's Law

At a given temperature, the product of pressure x volume for a gas is constant

$$P_1 V_1 = P_2 V_2$$

The pressure and volume of a gas are inversely proportional, provided the temperature and mass are kept constant



High pressure => Low volume

Low pressure => High volume

Dalton's Law

The partial pressure of a gas in a mixture of gases is the pressure that gas would exert if it occupied the total volume of the mixture. Thus, **partial pressure of a gas (P_x)** is the **total pressure multiplied by the fractional concentration** of dry gas.

$$P_x = P_B \times F$$

Corrected for water vapor: **$P_x = (P_B - P_{H_2O}) \times F$**

P_x = Partial pressure of a gas (mm Hg)

P_B = Barometric pressure (mm Hg)

P_{H_2O} = Water vapor pressure at 37°C (47mm Hg)

F = Fractional concentration of the gas in question (no units)

Consequently, **the sum of partial pressures of all gases in a mixture equals the total pressure of the mixture.**

Henry's Law for Concentrations of Dissolved Gases

Henry's law is used to **convert the partial pressure** of gas in the liquid phase to the **concentration of gas** in the liquid phase (e.g., in blood).

$$C_x = P_x \times \text{Solubility}$$

[%] **or** [mL gas/100 mL blood]

C_x = Concentration of dissolved gas (mL gas/100mL blood)

P_x = Partial pressure of gas (mm Hg)

Solubility = Solubility of gas in blood (mL gas/100mLblood per mm Hg)

Only applicable to **gases free in solution, NOT bound** - e.g., to protein such as hemoglobin

Fick's Law - Diffusion of Gases

$$V_x = \frac{DA\Delta P}{\Delta x}$$

V_x = Volume of gas transferred per unit time

D = Diffusion coefficient of the gas

A = Surface area

ΔP = Partial pressure difference of the gas

Δx = Thickness of the membrane

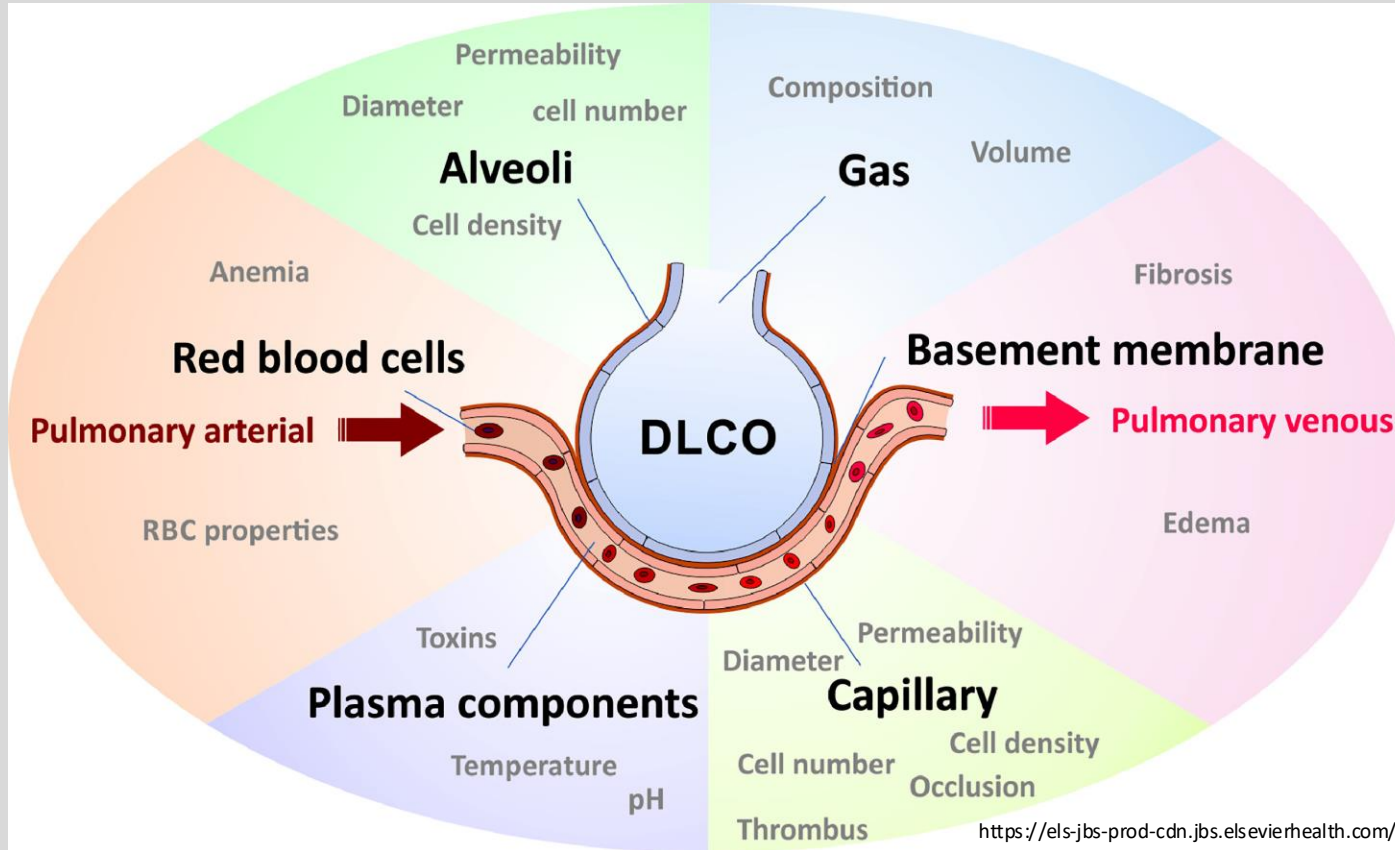
$$D_{CO_2} \gg D_{O_2}$$

Lung diffusing capacity (DL): Combines D , A , ΔX . Considers gases binding to proteins.

Lung Diffusion Capacity (DL)

How well oxygen gets into the blood from the air you breathe

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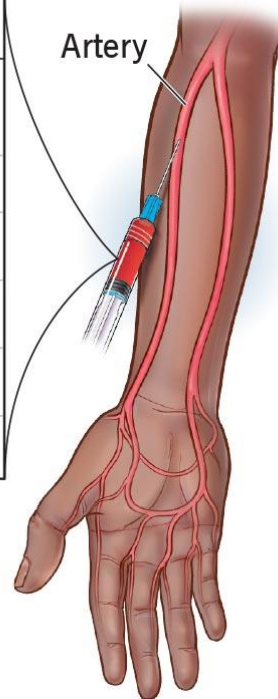
Forms of Gases in Blood

Oxygen
content

Oxygenated
hemoglobin

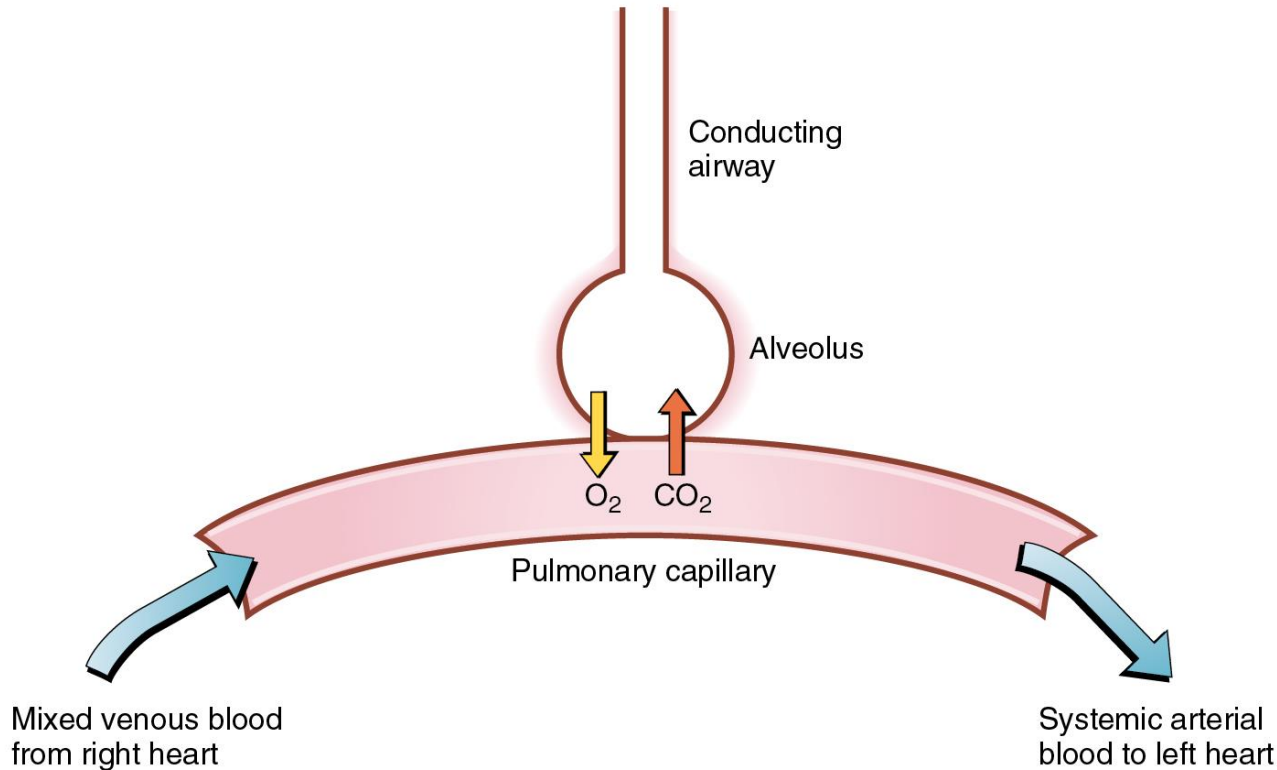
Arterial Blood Gas (ABG)

ABG	Normal range
O ₂ CT	15-23% per 100 mL of blood
pH	7.35-7.45
PaCO ₂	35-45 mmHg
PaO ₂	80-100 mmHg
HCO ₃	22-26 mEq/L
O ₂ Sat	95-100%



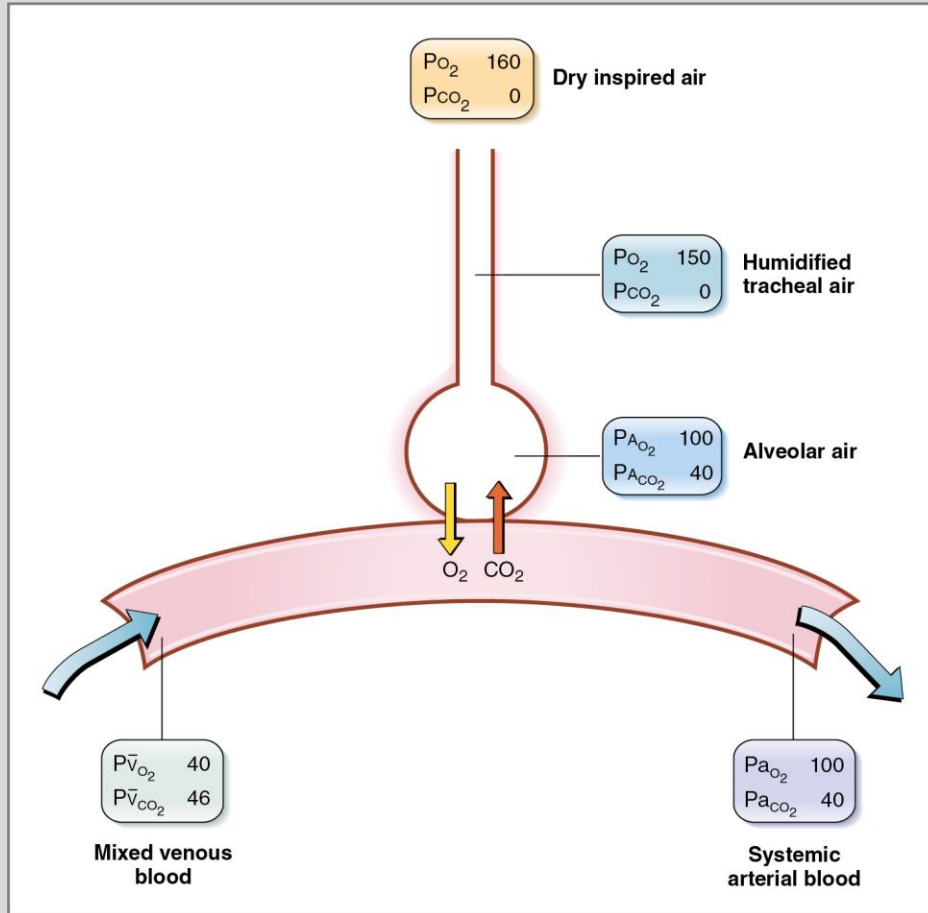
Dissolved, bound, chemically modified

Gas Exchange in the Lungs



Gas Exchange in the Lungs Continued

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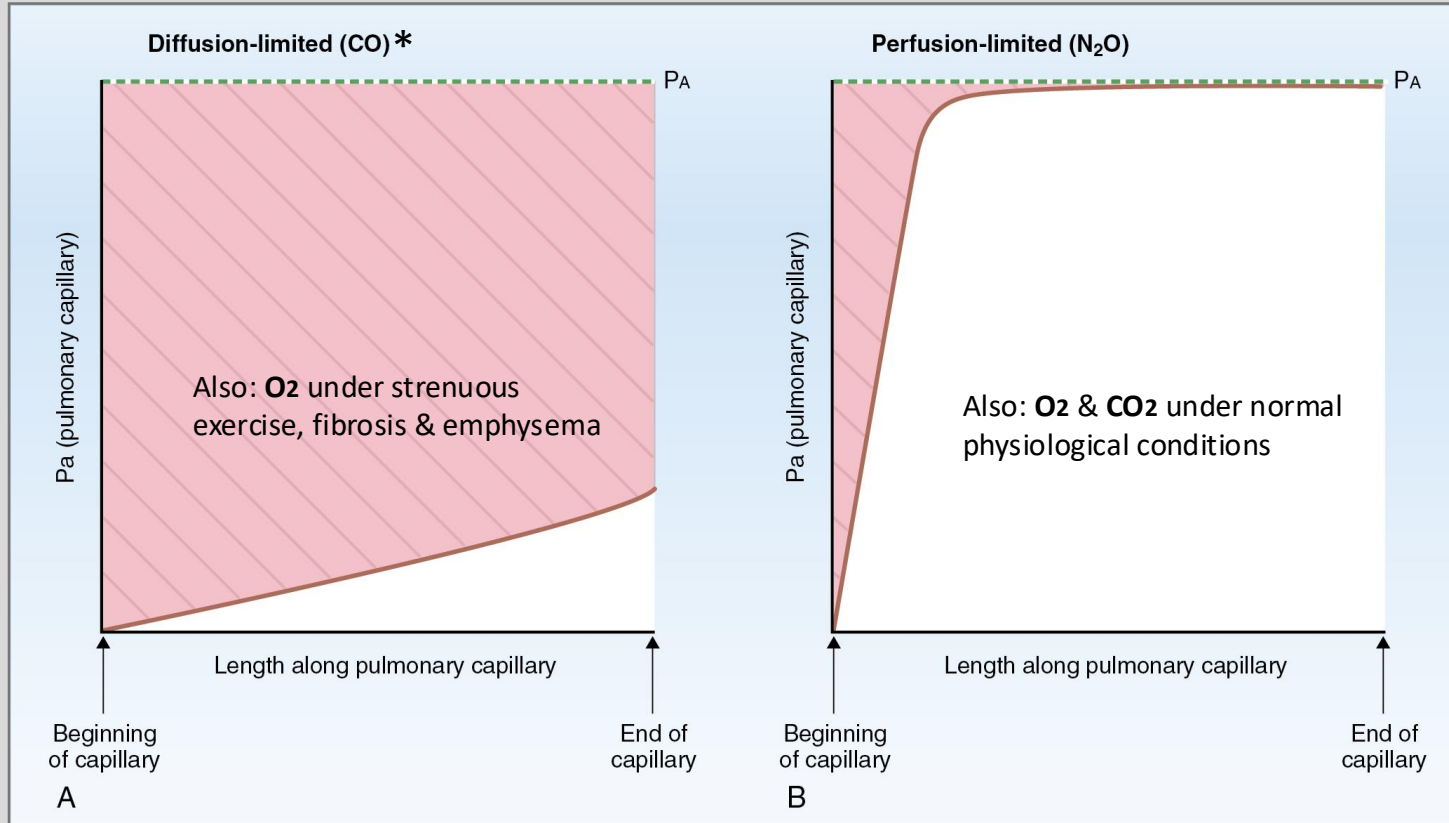


The P_{O_2} of humidified tracheal air is 150 mm Hg ($[760 \text{ mm Hg} - 47 \text{ mm Hg}] \times 0.21 = 150 \text{ mm Hg}$)

A/a - Ventilation/perfusion defect increases with certain pathologies

Bronchial blood flow and a small portion of coronary venous blood

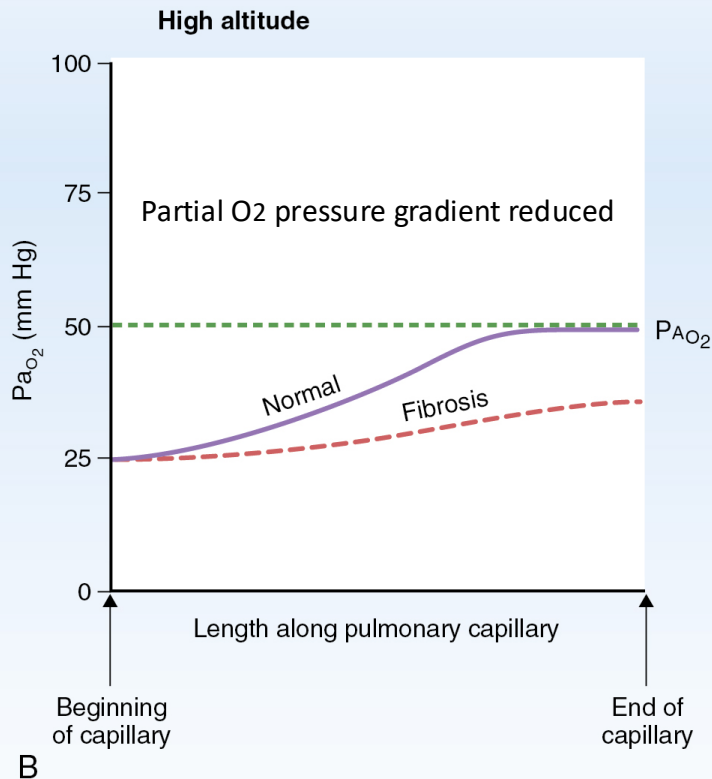
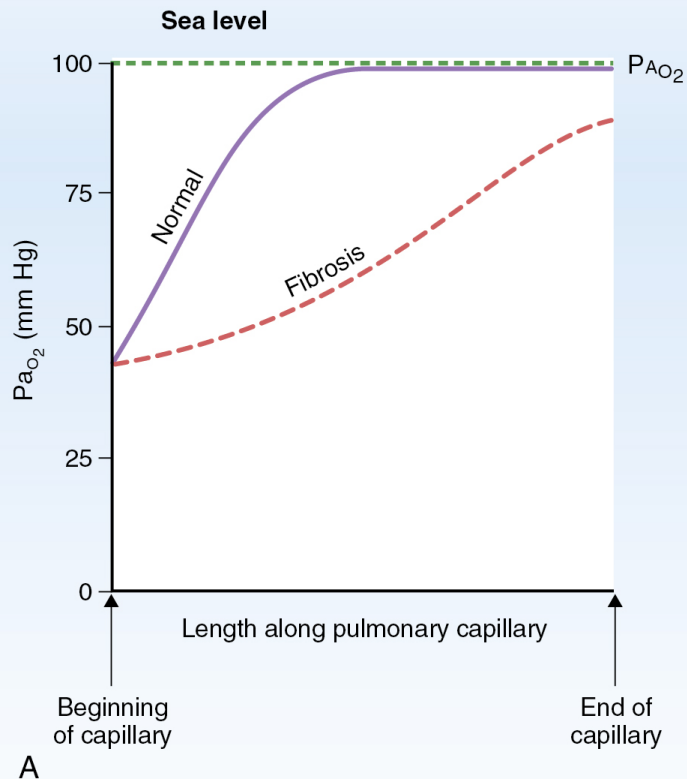
Diffusion-limited & Perfusion-limited Gas Exchange



*Most CO not free as it immediately binds to Hgb $\rightarrow$ Does not partake in pressure gradient

O₂ Diffusion Limited by Pathology & High Altitude

O₂ DIFFUSION



Practice Medical Board-Style Question

A clinical physiology researcher is testing alveolar gas equilibration in a healthy 25-year-old athlete performing maximum-intensity treadmill exercise. During intense exercise, the velocity of cardiac output increases significantly, reducing the transit time of an individual erythrocyte through the alveolar capillary from a resting time of 0.75 seconds down to a challenge transit time of just 0.25 seconds. Under **normal physiological conditions**, how does this shortened transit time affect the capillary equilibration of oxygen (O_2), and what type of transport limit does it demonstrate?

- A. O_2 transport becomes diffusion-limited because the shortened transit time prevents the partial pressure of capillary O_2 from matching alveolar levels.
- B. O_2 transport becomes perfusion-limited because partial pressure equilibration is reached well within the shortened 0.25-second window.
- C. O_2 transport switches to being diffusion-limited because carbon dioxide (CO_2) competes for the same capillary surface area during intense exercise.
- D. O_2 transport becomes perfusion-limited because the high velocity of blood flow prevents any gas diffusion from taking place across the blood-air barrier.
- E. O_2 transport remains diffusion-limited because the thickening of the blood-air barrier during exercise slows down the physical movement of gas molecules.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

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